# RTL_EXERCISE_1 RING FLASHER  EX

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## Contents

1. Interface ..... 2

2. Functional implementation. ..... 4

3. Internal implementation. ..... 10

3.1. Overall. ..... 10

3.2. State Machine ..... 11

4. History ..... 13

## 1. Interface

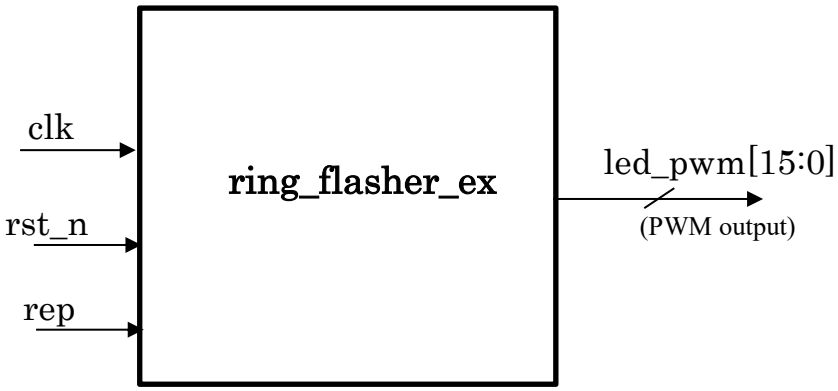



Figure 1: the figure of Ring Flasher EX System

| Signal | Width | In/Out | Description |
|---|---|---|---|
| clk | 1 | In | System clock input |
| rst_n | 1 | In | Active low, asyschronous reset |
| rep | 1 | In | Repeat signal. When active at final ACW step, the movement cycles repeats. When inactive, LEDs decay then system returns to IDLE. |
| led_pwm | 16 | Out | PWM output for each LED. Each bit corresponds to one physical LED on the ring. |

Table 1.1: Description of signals in Ring Flasher EX

| Parameter | Default | Description |
|---|---|---|
| NUM_LEDS | 16 | Number of LEDs in the ring. |
| CW_STEPS | 12 | Number of clockwise steps per movement cycle. |
| ACW_STEPS | 8 | Number of anti-clockwise steps per movement cycle. |
| STEP_TICKS | 2500 | Clock cycles between each step (controls animation speed). |
| NUM_LEVELS | 6 | Number of brightness levels (0 = off, |

| | | |
|---|---|---|
| | | NUM_LEVELS - 1 = Maximum |
| STEP_DECAYS | 2500 | Clock cycles between each brightness decay step in DECAY state |

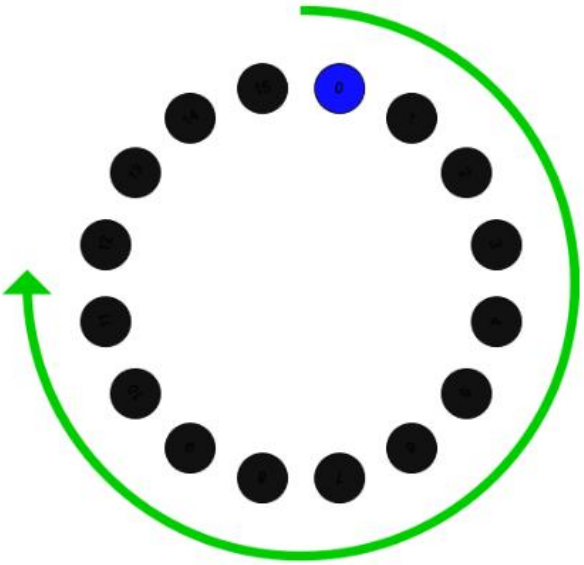
Table 1.2: Description of parameter in Ring Flasher EX

## 2. Functional implementation.

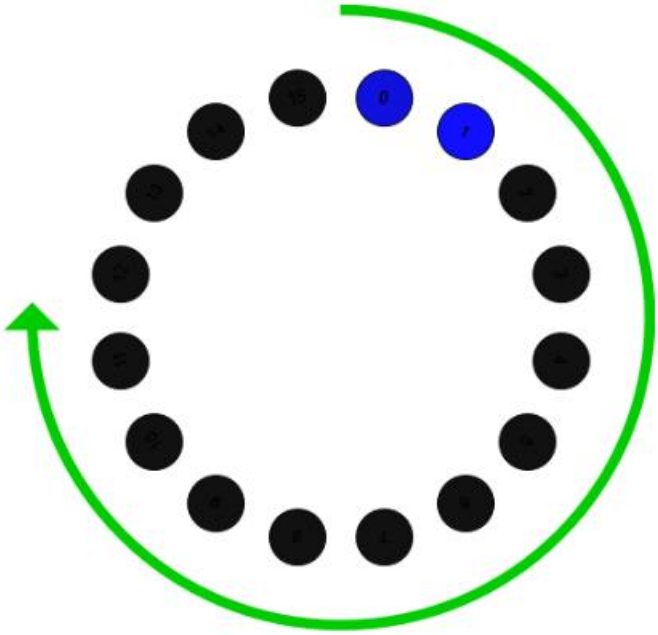
- Implement The ring_flasher_ex module drives a ring of NUM_LEDS LEDs with PWM brightness control. Each LED can be at one of NUM_LEVELS brigthness levels (0 = off, NUM_LEVELS = fully on). The system behaviour is defined by the following rules:
- System's Operation base on three input signal
  - Reset
  - Clock
  - Rep
- The system specification
  - Clock signal is provided for system inspire of function status. The function operate state's transition at positive edge of the clock signal.
  - Reset signal:
    - LOW-ACTIVE Reset = 0: all brightness registers clear to 0, state returns to IDLE, head_pos and step_cnt reset to 0.
    - HIGH-ACTIVE Reset = 1: System is started with IDLE state.
- Movement cycle
  - A single movement cycle consists of CW_STEPS clockwise (CW) steps followed by ACW_STEPS anti-clockwise (ACW) steps.
  - Each step is separated by STEP_TICKS clock cycles.
  - At every step, the head LED is set to maximum brightness (NUM_LEVELS – 1). All orther lit LEDs have their brightness decrement by 1 (trail effect).
- Repeat behaviour
  - At the final ACW step of a movement cycle, the rep input is sampled:
    - If rep = 1: a new movement cycle starts immediately (CW phase again).
    - If rep = 0: the system enters DECAY state. No further steps are taken.
    - Asserting rep after the final ACW step has already been evaluate has no effect – the system remain in DECAY.
- DECAY and return to IDLE
  - In DECAY state, all active LEDs decrease in brightness by 1 level every STEP_DECAYS clock cycles.
  - When all LEDs reach brightness 0, the system returns to IDLE.

- PWM output
  - A free-running counter pwm_cnt cycles from 0 to NUM_LEVELS−2.
  - led_pwm[i] = 1 when brightness[i] > pwm_cnt, producing a duty cycle proportional to brightness level.
  - Brightness level 0 → 0 % duty cycle (LED off), level NUM_LEVELS−1 → 100 % duty cycle (LED fully on).
- Some insulations:
  - At the initial IDLE state, all LEDs are off.

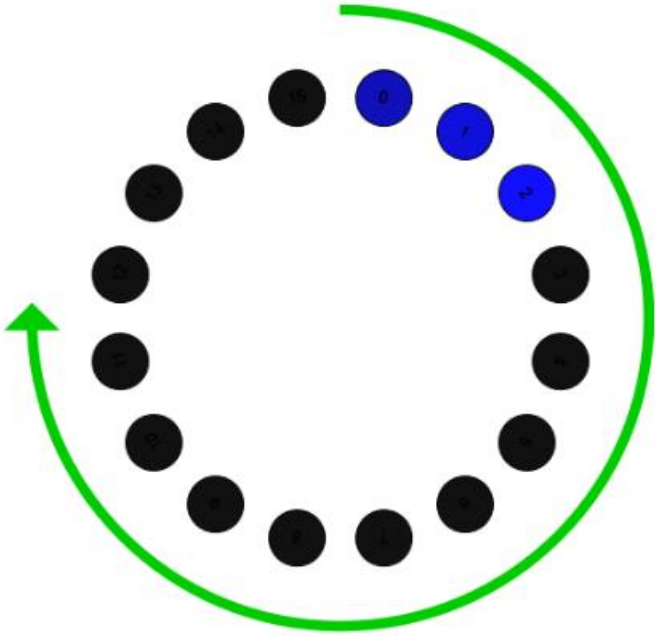
  - If the active-HIGH rep signal is active while the module is in its idle state, the flasher will start operating.



- At the first step, led[0] will turn on at maximum brightness.
- 0.5 second later, the module moves on to led[1]. led[1] will now turn on at maximum brightness, while led[0]'s brightness level will be reduced by 1.

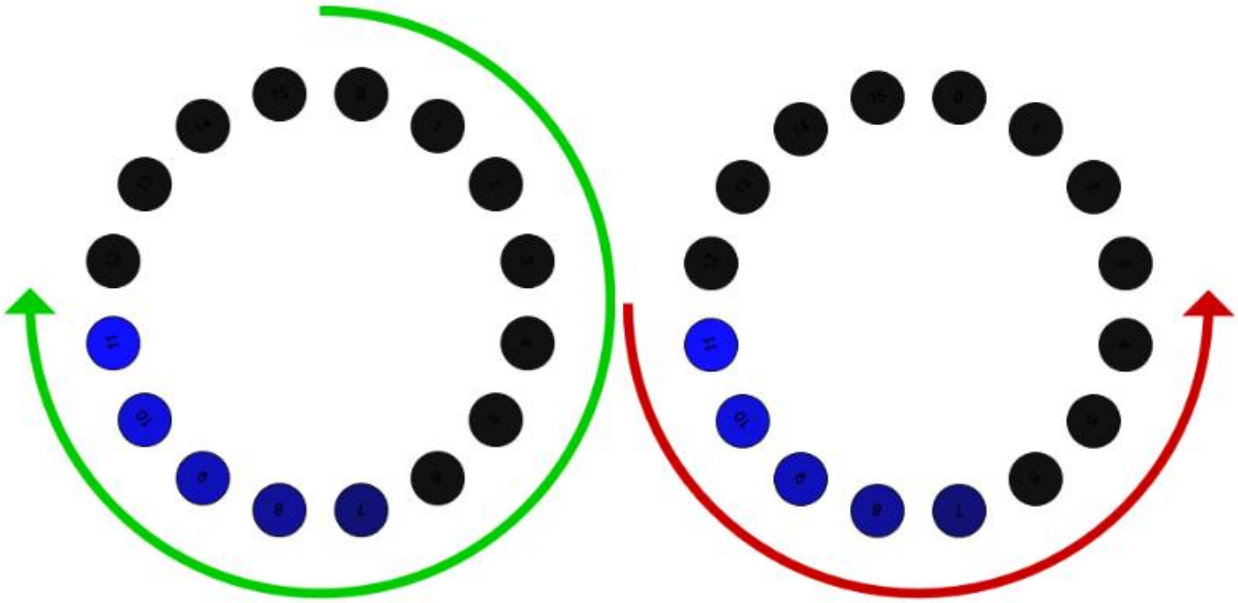


- Another 0.5 second later, the module moves one step clockwise (CW) to led[2]. led[2] will similarly be turned on at max brightness, while led[0] and led[1] will have their brightness level each be decreased by 1.

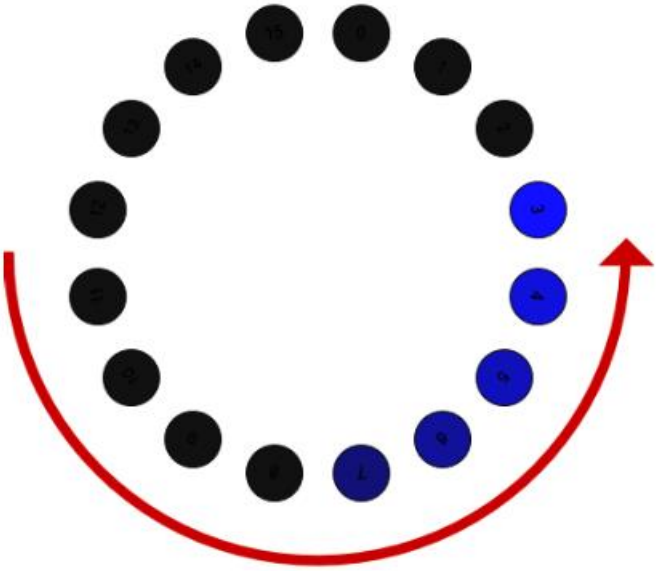


- This process repeats every 0.5s. Note that an LED's brightness level cannot go below 0.

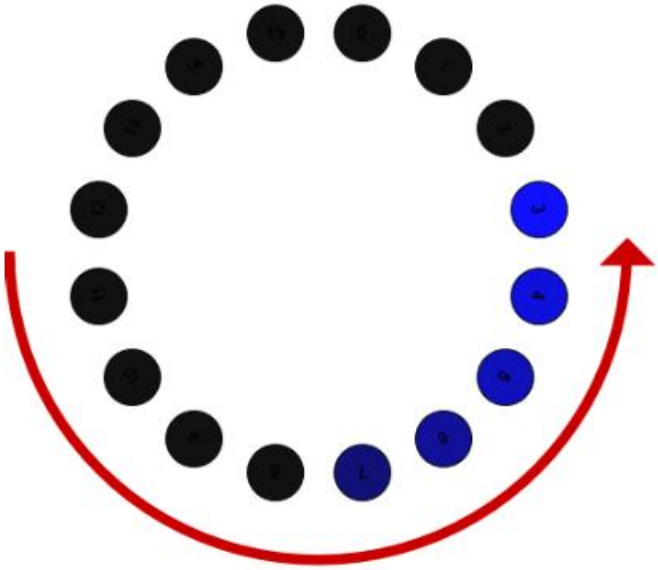
- After 5.5s (i.e. when 12 CW steps have been taken in total), the module will change direction to move in the anti-clockwise (ACW) direction.



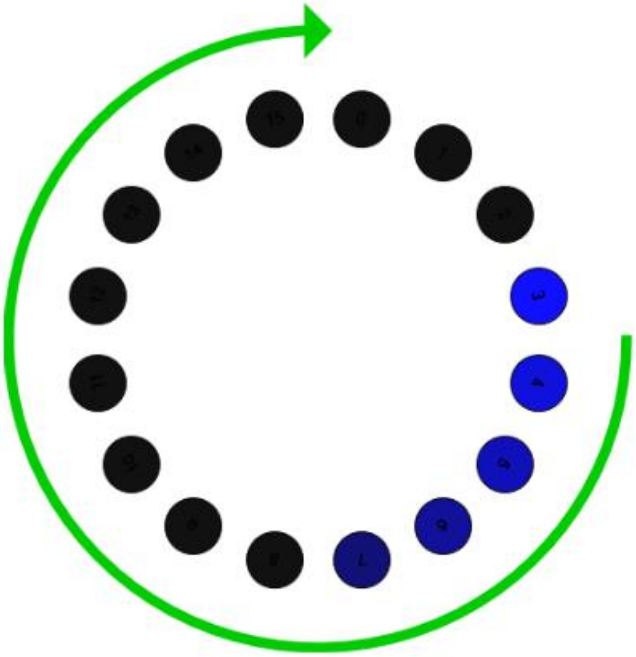
- 0.5s later, the first ACW step is taken. led[10] is now at max brightness, and all other LEDs decay in brightness like normal.



- This repeat every 0.5s util 8 ACW steps have been taken in total.



- At which point the module will change direction again, taking another 12 CW step every 0.5s.



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- Then the module will once again change direction to take 8 more ACW steps.
- This combination of 12 CW steps followed by 8 ACW steps is henceforth referred to as the module's **movement cycle**.
- The module will keep repeating movement cycles so long as the *rep* signal is active at the final ACW step of a movement cycle. Otherwise, no further steps are taken and the

current active LEDs will simply decay in brightness at set intervals of 0.5s (reactivating *rep* at this point does nothing). The module will return to the initial idle state when all LEDs are at brightness level 0.

## 3. Internal implementation.

3.1. Overall.

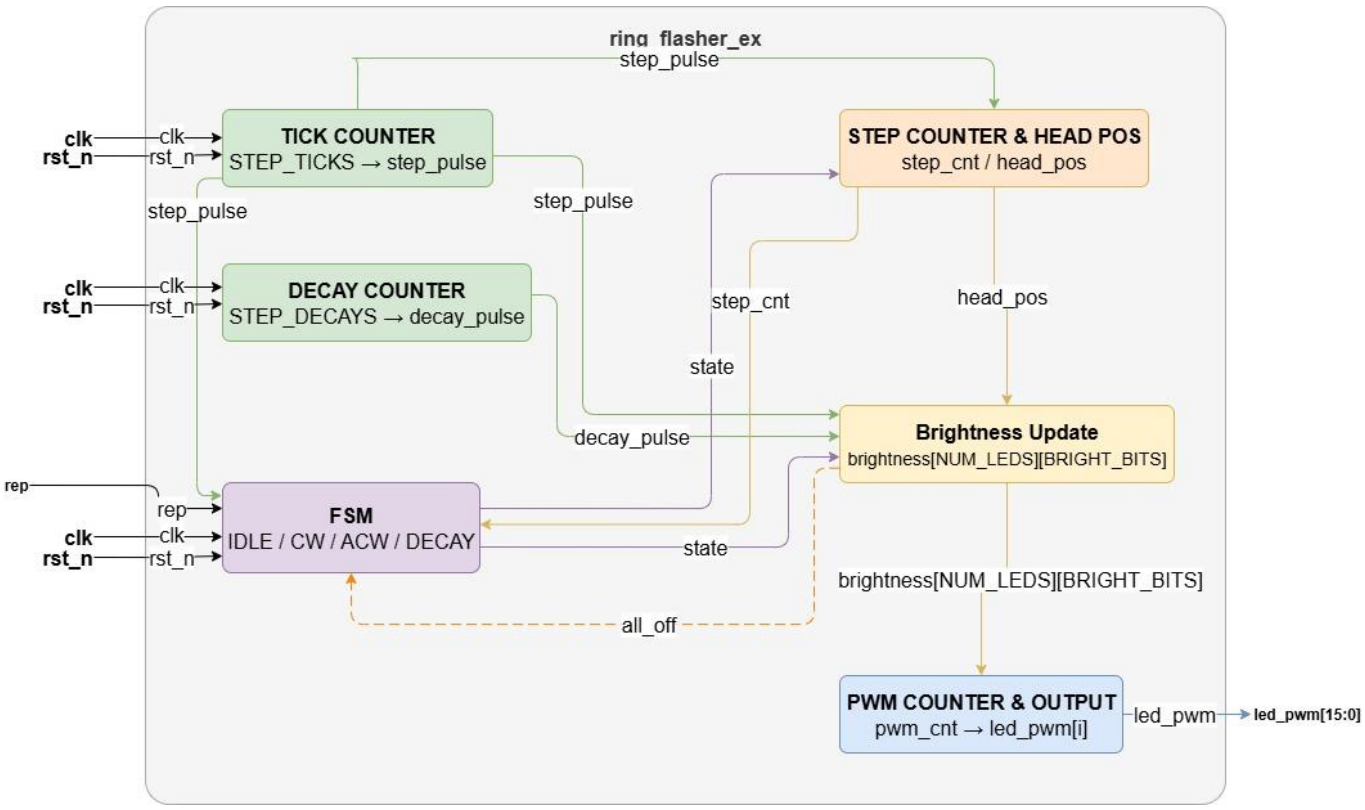


Figure 3.1: Block diagram of Ring Flasher EX

| Functional Block / Signal | Description |
|---|---|
| PWM COUNTER & OUTPUT | Generates per-LED PWM signals from brightness registers. |
| TICK COUNTER | Produces step_pulse every STEP_TICKS cycles during CW/ACW. |
| DECAY COUNTER | Produces decay_pulse every STEP_DECAYS cycles during DECAY. |
| FSM | Controls state transitions based on step_pulse, step_cnt, rep, and all_off. |
| STEP COUNTER & HEAD POSITION | Tracks current step and active LED index. |
| BRIGHTNESS UPDATE | Updates brightness array on each step_pulse or decay_pulse. |
| brightness[N][BRIGHT_BITS] | Packed 2D array holding the current brightness level (0 to NUM_LEVELS-1) of each LED. |
| pwm_cnt | Free-running counter 0 to NUM_LEVELS-2. Compared with |

| | |
|---|---|
| | brightness[i] to produce PWM output. |
| tick_cnt | Counter for step timing. Counts STEP_TICKS cycles, active during CW and ACW states. |
| decay_cnt | Counter for decay timing. Counts STEP_DECAYS cycles, active during DECAY state. |
| step_cnt | Step counter. Counts steps within current CW or ACW phase, resets on state transition. |
| head_pos | Index (0 to NUM_LEDS-1) of the currently active LED. Increments in CW, decrements in ACW. |
| step_pulse | Single-cycle pulse generated when tick_cnt reaches STEP_TICKS-1 in CW or ACW state. |
| decay_pulse | Single-cycle pulse generated when decay_cnt reaches STEP_DECAYS-1 in DECAY state. |
| all_off | Combinational flag, asserted when all brightness[i] equal 0. Triggers DECAY-to-IDLE transition. |

Table 3.1: Block diagram of Ring Flasher EX Description

3.2. State Machine

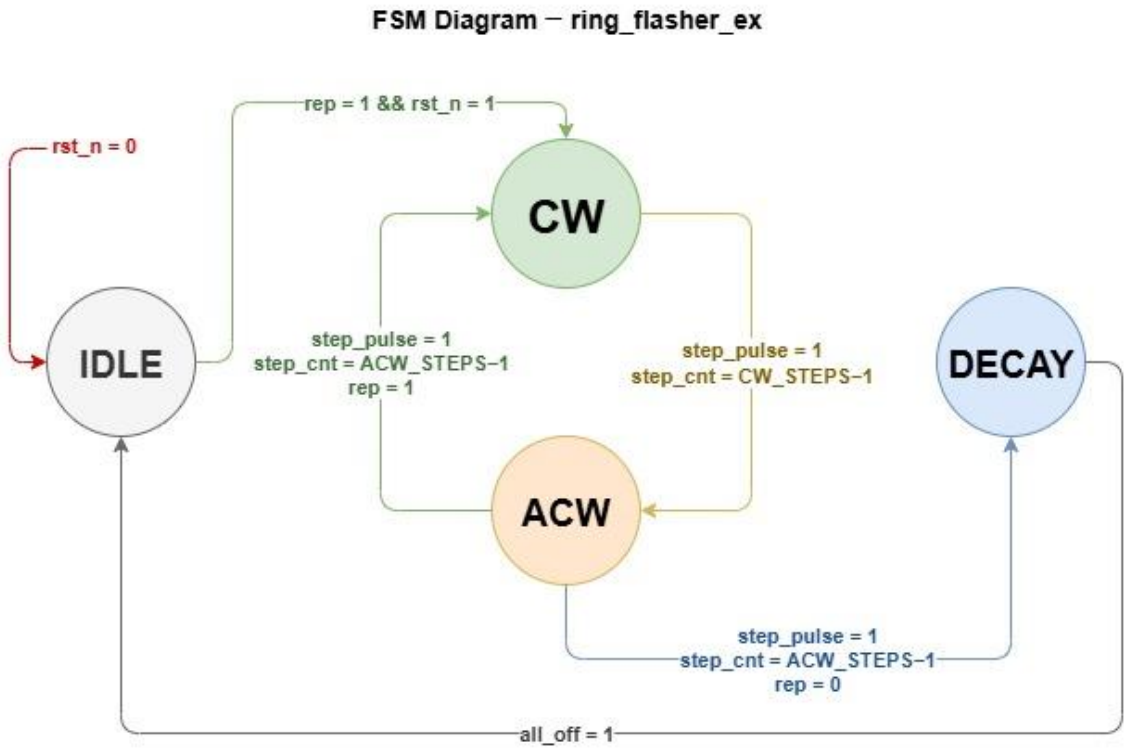

Figure 3.2: State Machine of Ring Flasher EX

| Variable name | Witdh | Datatype | Description |
|---|---|---|---|
| state | 2 | enum logic | Current state (IDLE/CW/ACW/DECAY) |
| next | 2 | enum logic | Next state (IDLE/CW/ACW/DECAY) |

Table 3.2: variable name of State machine

| State | Encoding | Description |
|---|---|---|
| IDLE | 2'd0 | Initial/reset state. All LEDs are off. Waits for rep=1 to start a movement cycle. |
| CW | 2'd2 | Clockwise stepping. Head LED advances CW_STEPS positions around the ring, each step separated by STEP_TICKS clock cycles. |
| ACW | 2'd1 | Anti-clockwise stepping. Head LED retreats ACW_STEPS positions. At the final ACW step: if rep=1 returns to CW, otherwise goes to DECAY. |
| DECAY | 2'd3 | All active LEDs decay in brightness by 1 level every STEP_DECAYS cycles. Returns to IDLE when all LEDs reach brightness 0. |

Table 3.3: state name of State machine

## 4. History

| Date | Author | Modified part | Description |
|---|---|---|---|
| 2026/03/31 | Group 2 | All | New creation |
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